

The Credentials Arms Race: Understanding Immigrant Over-Education in the U.S. Labor Market

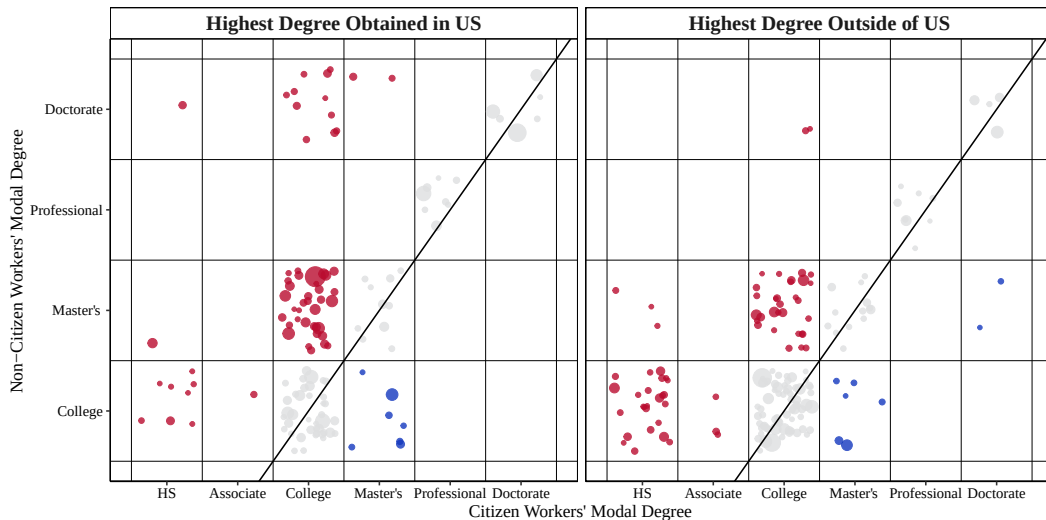
Willy Chen

Midwest Economics Association Annual Meeting
Mar. 21, 2026



Motivation

Figure 1: Educational Mismatch of Non-Citizens, ACS 2010-2019



Related Literature

This paper concerns four strands of literature:

- ① Over-education and Wage Penalty (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).

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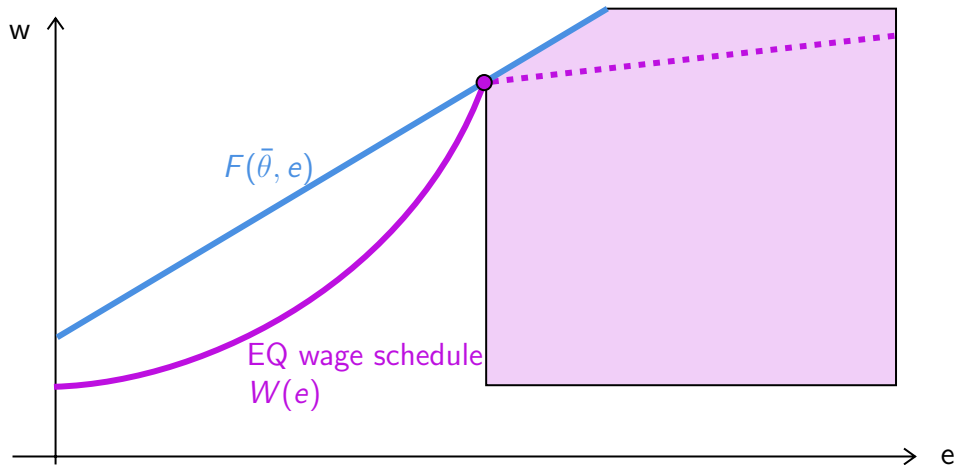
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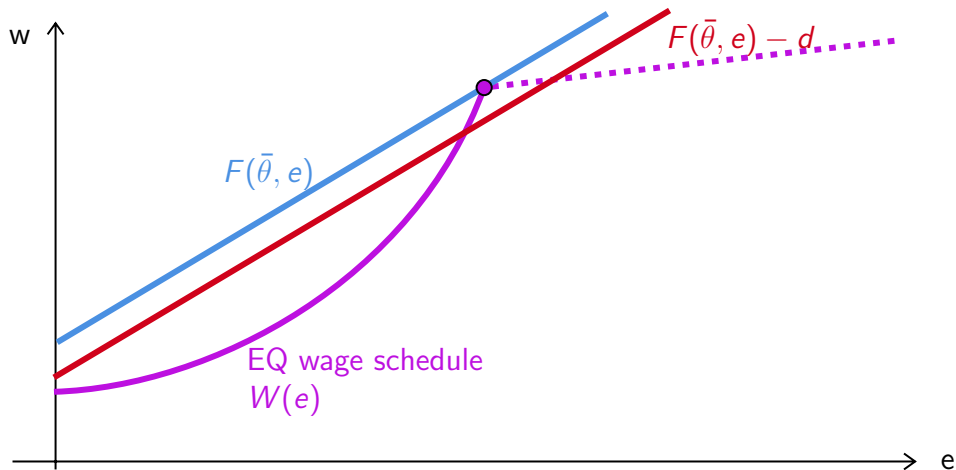
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- ④ Separating equilibria in games with productive signals (Chatterjee et al., 2021; Cho et al., 1987; Kaymak, 2025; Mailath, 1987; Spence, 1978; Tyler et al., 2000).

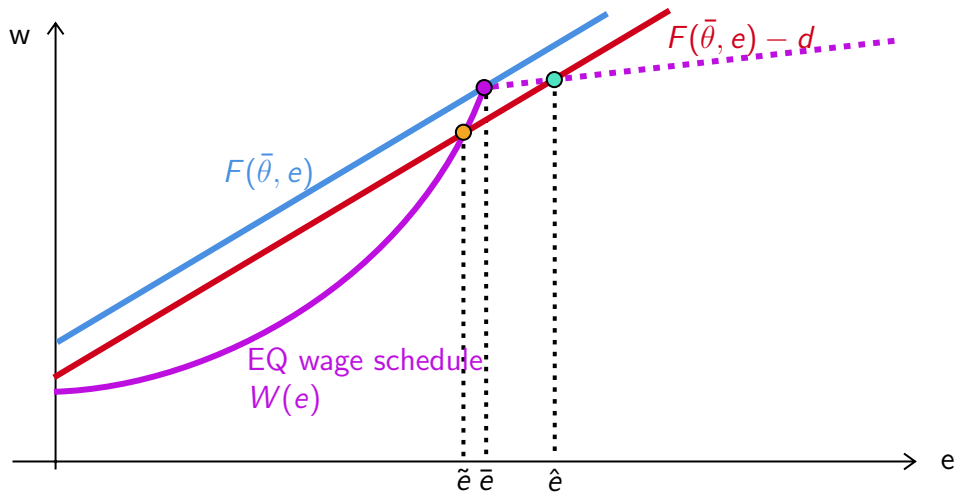
Model: Productive Signals with Continuous Types



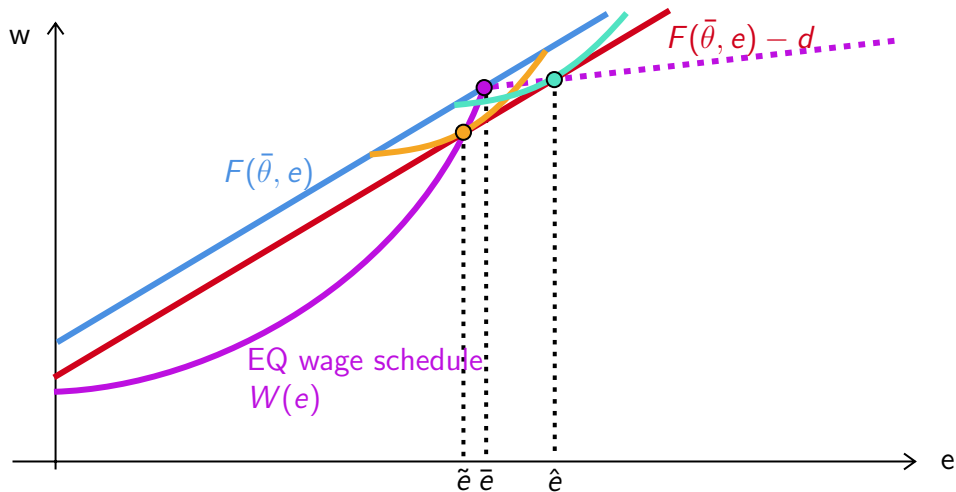
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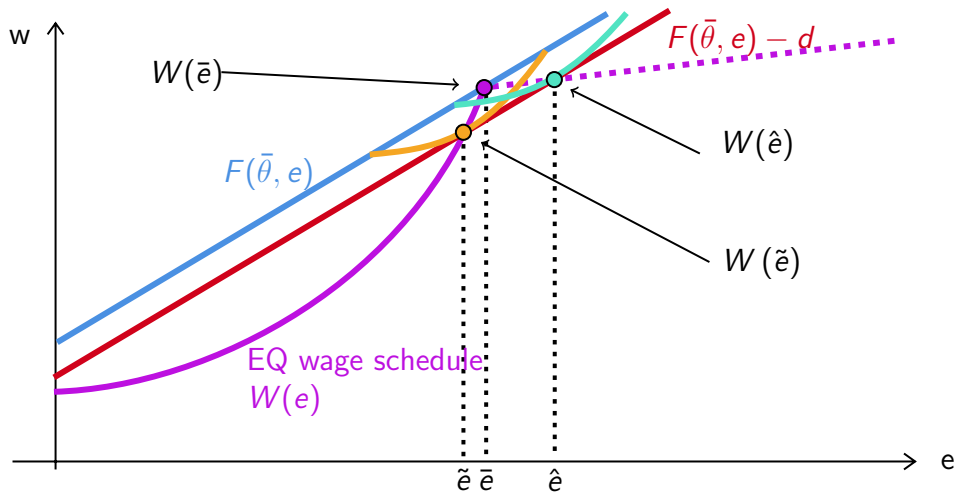
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$$d^s = W(\bar{e}) - \left[E[W(\bar{e} + \alpha)] + \frac{E[W(\bar{e} + \beta)] - E[W(\bar{e} + \alpha)]}{E[\beta - \alpha]} \cdot E[-\alpha] \right].$$

Employer learning implies that $\alpha, \beta \rightarrow 0$ over time.

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The cost gap observed right after graduation is $d = d^t + d^s$.

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By comparing within-cohort observed d over time, I can bound d^t and d^s .

Controlling for Job Characteristics

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② Importance of Certification vs. Typical On-the-Job Training [Graphs](#)

LinkedIn Profiles

To study whether employer learning reduces education–occupation gaps—something cross-sectional data (e.g., ACS) cannot address—I construct a panel using scraped LinkedIn profiles from People Data Labs.

Current data include:

- Person record (sex, job, location)
- Job experience (date, company, title, role)
- Education (date, school, degree, major)
- Certifications (date, organization)

Matching LinkedIn Job Titles to SOC's

Using 250 distinct job titles for each of the 1,016 SOC codes from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

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Instruction: Map title to SOC

Input: Vice President, company_name

Response: 11-1011.00

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Instruction: Map title to SOC

Input: Cloud Tree Guitar

Response: 00-0000.00

Matching LinkedIn Titles to SOC

Table 1: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	O*NET SOC	Corresponding O*NET Title
director - field support	27-2012.05	Media Technical Directors/Managers
senior page designer and copy editor	27-3041.00	Editors
gflwn askdm	00-0000.00	NA
taylor mattis fellow	51-9195.05	Potters, Manufacturing
solar analyst intern	17-2199.11	Solar Energy Systems Engineers
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9032.00	Education Administrators, Kindergarten through Secondary
asset & reliability engineer	17-2199.00	NA
sales / acct manager	41-1011.00	First-Line Supervisors of Retail Sales Workers
attest intern	13-2099.00	NA
immersion associate	25-1194.00	Career/Technical Education Teachers, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers

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attest intern	13-2099.00	NA
immersion associate	25-1194.00	Career/Technical Education Teachers, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers
網頁工程師	15-1299.08	Computer Systems Engineers/Architects
博士生	25-9044.00	Teaching Assistants, Postsecondary

LinkedIn Panel

I construct a panel of individuals who attended a U.S. degree-granting post-secondary institution and held at least one job matched to an O*NET SOC code. The sample spans 1970–2025 and includes $\sim 1\text{B}$ person-year observations from $\sim 53\text{M}$ individuals, averaging 18.6 years per person.

Table 2: Combined Education and Employment Statistics

Part 1: By Highest Degree			Part 2: Education-to-Employment Gap		
% Degree	LinkedIn	ACS	Gap Duration	Count (N)	Pct. %
Assc & Some College	34.8	41.6	Same year (0)	31,339,560	44.6
Bachelor's	36.7	37.5	1 year	14,087,376	20.1
Master's	22.8	15.2	2 years	6,999,362	10.0
Professional	3.2	3.4	3–5 years	10,814,281	15.4
PhD	2.5	2.3	> 5 years	7,015,214	10.0

Constructing Smooth Education Attainment

Using IPEDS data, I can observe variation in the prestige of schools where each worker acquired their degree.

$$\text{score}_s = 0.65 \times \text{Std. SAT/ACT Scores} + 0.35 \times \text{Std. Rejection Rate}$$

At each degree level, I can then apply the calculated score to proxy for the signaling value of the degree.

$$\mathbb{I}_i = \beta_0 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \delta_d \cdot \mathbb{1}\{\text{Deg}_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \beta_d \cdot \mathbb{1}\{\text{Deg}_i = d\} \cdot (\text{score}_i - \text{score}_d) + \varepsilon$$

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$$\hat{e}_i = 1 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \hat{\delta}_d \cdot \mathbb{1}\{\text{Deg}_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \hat{\beta}_d \cdot \mathbb{1}\{\text{Deg}_i = d\} \cdot (\text{score}_i - \text{score}_d)$$

Estimated Education Signals

Table 3: Estimated Marginal Signaling Value of Degrees using Nakao-Treas Prestige Score

Tier	School	Bachelors	Masters	PhD	Professional
2	Bottom 25%	-2.14	0.96	2.92	9.45
3	Top 75%	-1.92	1.11	3.26	10.18
4	Top 50%	-1.54	1.58	3.52	10.67
5	Top 33%	-1.33	1.54	3.28	11.07
6	Top 20%	-0.62	1.88	3.84	11.04
7	Top 10%	-	2.60	4.19	10.75
8	Top 5%	1.54	3.12	4.22	9.64
9	Top 1%	2.88	3.27	4.39	8.34

Future Steps

- Fully implement the construction of $\hat{e}$ to estimate the equilibrium wage function and use it to identify d .

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- Use the full panel to separate d^t from d^s .
- Counterfactual analysis with increases in d —The new \$100,000 H-1B fee.

Questions?

Thank You!

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General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

Each worker chooses their education according to $\sigma : \Theta \rightarrow \mathcal{E}$.

Observing e , the firm updates belief about the worker's type $\hat{\Theta}(e) \subseteq \Theta$.

In a full **separating** equilibrium, σ is a one-to-one function, i.e.,

$$\hat{\Theta}(e) = \{\sigma^{-1}(e)\}, \forall e \in \mathcal{E} \quad (1)$$

and $\sigma(\theta)$ must satisfy *strict incentive compatibility*:

$$\{\sigma(\theta)\} = \underset{e}{argmax} U(\theta, \sigma^{-1}(e), e), \forall \theta \in \Theta \quad (2)$$

Regularity Conditions

- ① (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
- ② (Belief Monotonicity) $\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e) > 0$.
- ③ (Type monotonicity) $\frac{\partial^2}{\partial \theta \partial e} U(\theta, \hat{\theta}, e) > 0$
- ④ (“Strict” Quasi-Concavity) $\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) = 0$ has a unique solution in e , denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{\operatorname{argmax}} U(\theta, \hat{\theta}, e)$, and $\left. \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \right|_{e=\phi(\theta)} < 0$.
- ⑤ (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \Rightarrow \left| \frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) \right| > k.$$

Additionally, there are two niceness conditions: Initial Value (IV) and Single Crossing (SC).

Characterization of the Solution

Mailath (1987) shows that

If conditions 1-5 are satisfied, then a strictly increasing function σ , continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

① σ solves
$$\frac{d\sigma}{d\theta} = - \frac{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}$$

② when $\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e) > 0$, $\frac{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}$ is a strictly increasing function of θ .

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

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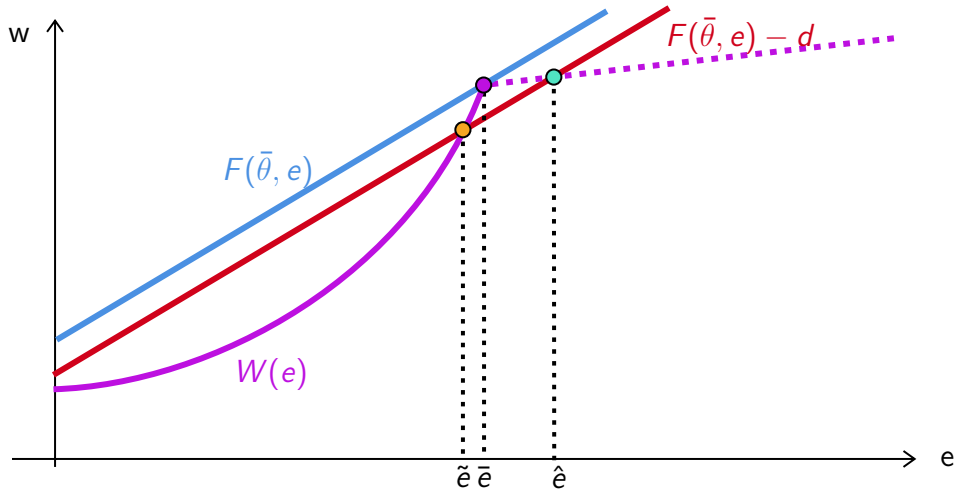
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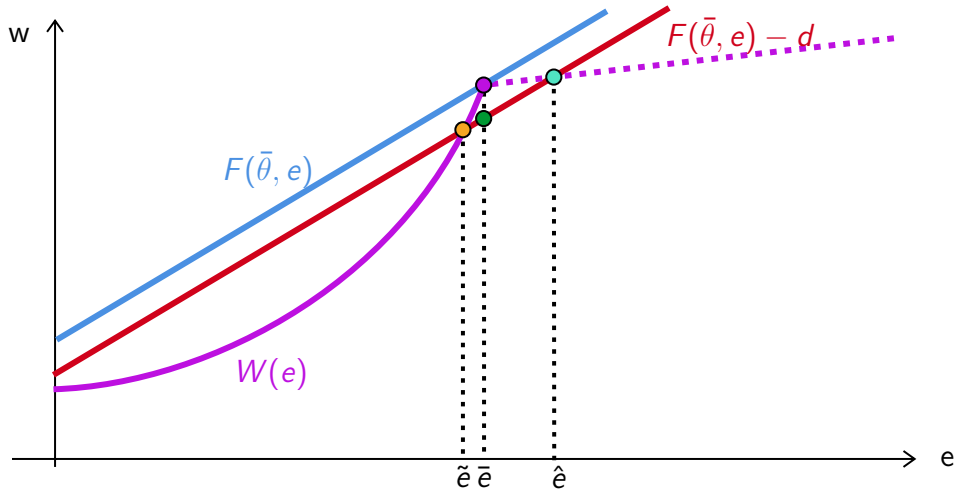
Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

In other words, there is a unique full separating equilibrium in the natives' labor market given the assumptions in the basic model environment.

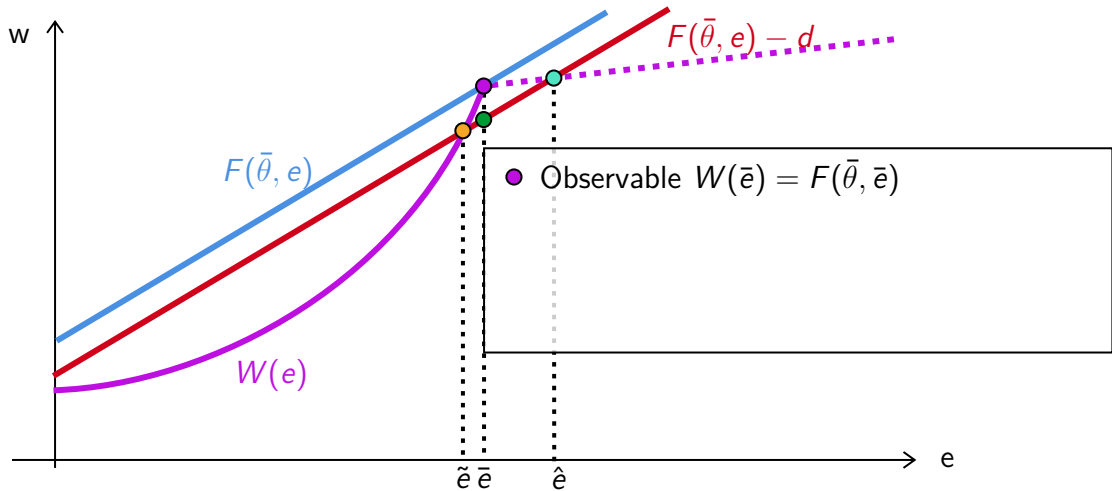
New Separating Equilibrium



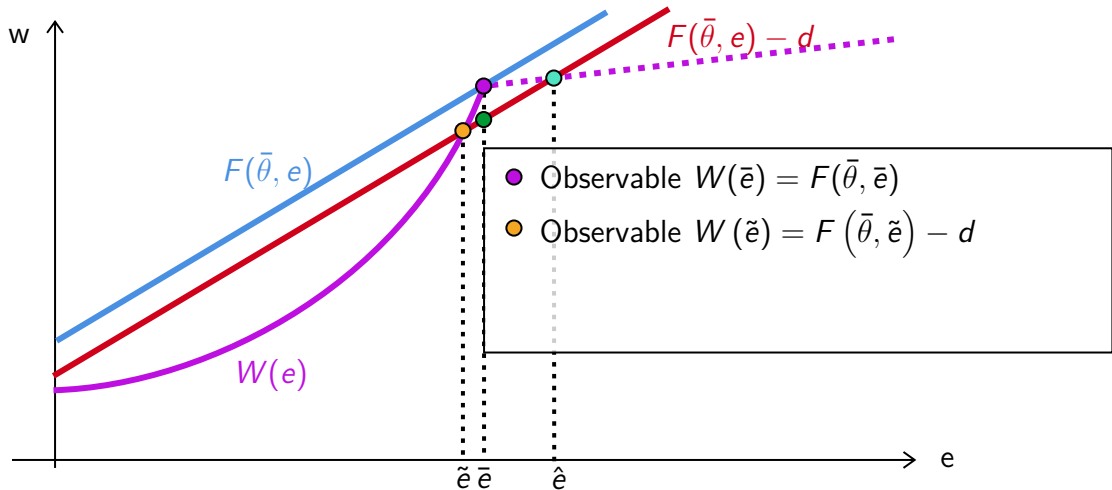
New Separating Equilibrium



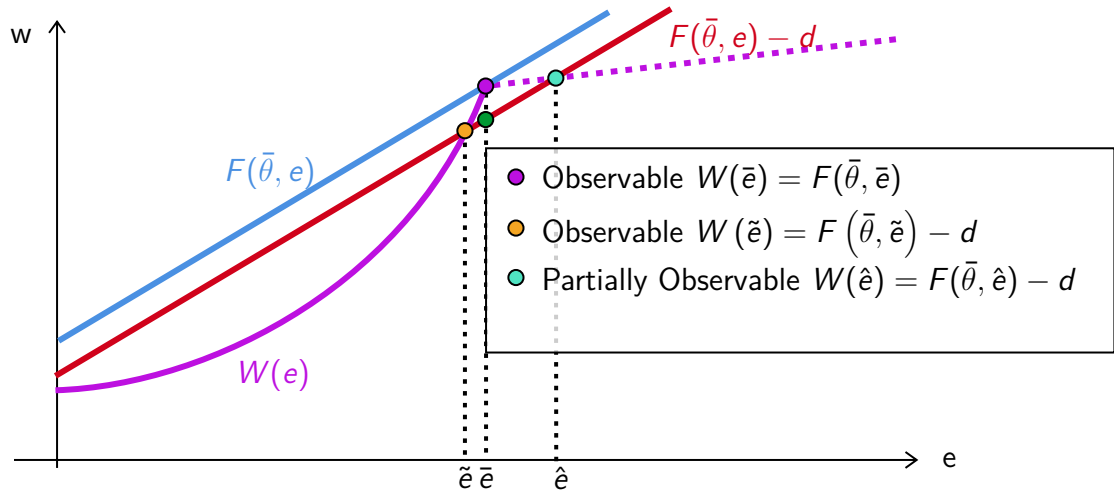
New Separating Equilibrium



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New Separating Equilibrium

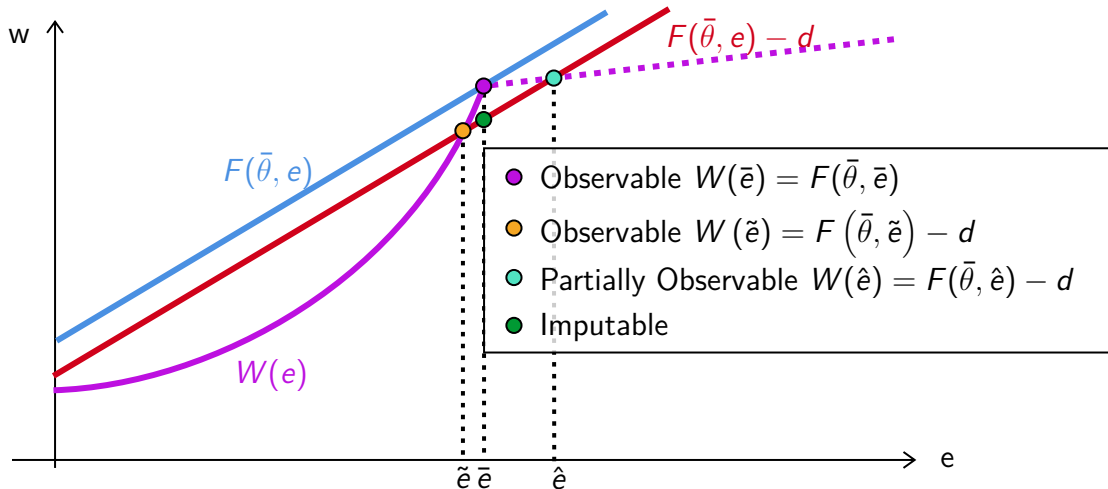
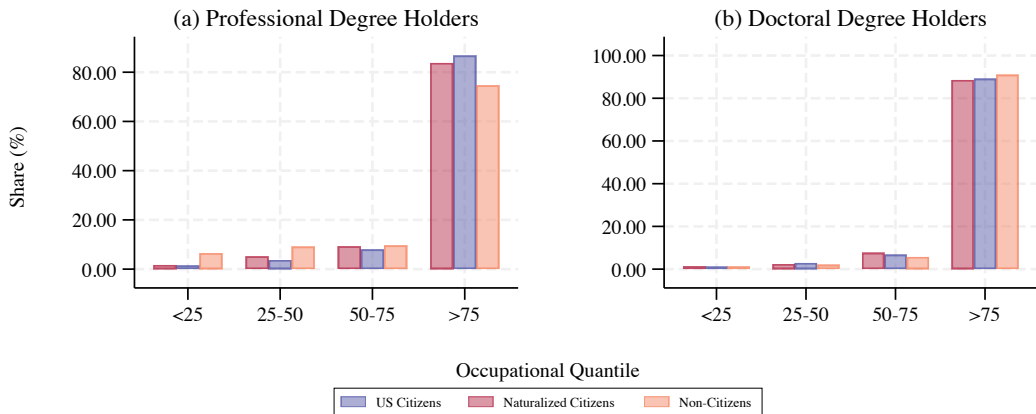
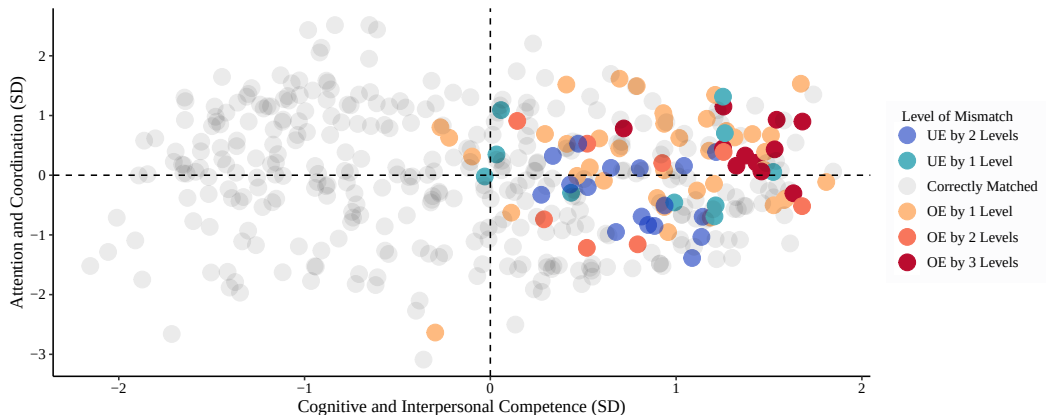


Figure A2: Educational Mismatch of Single Immigrants at High Degree Levels, 2010-2019



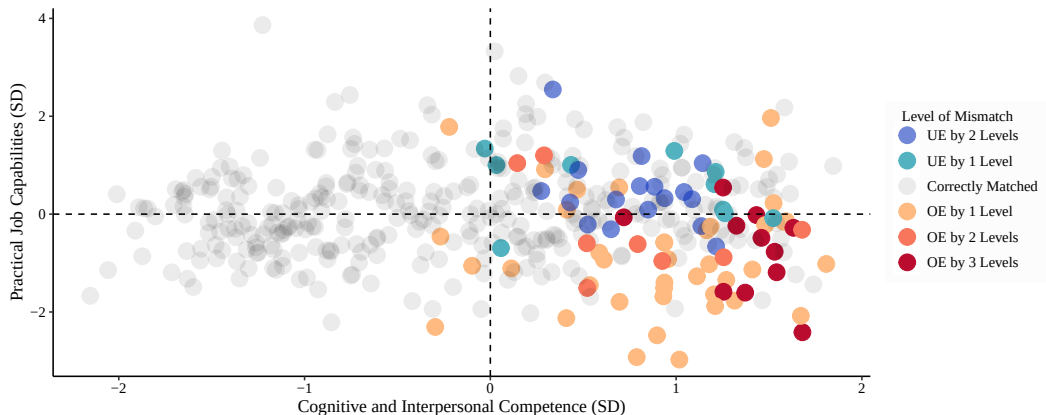
Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

Figure A3: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



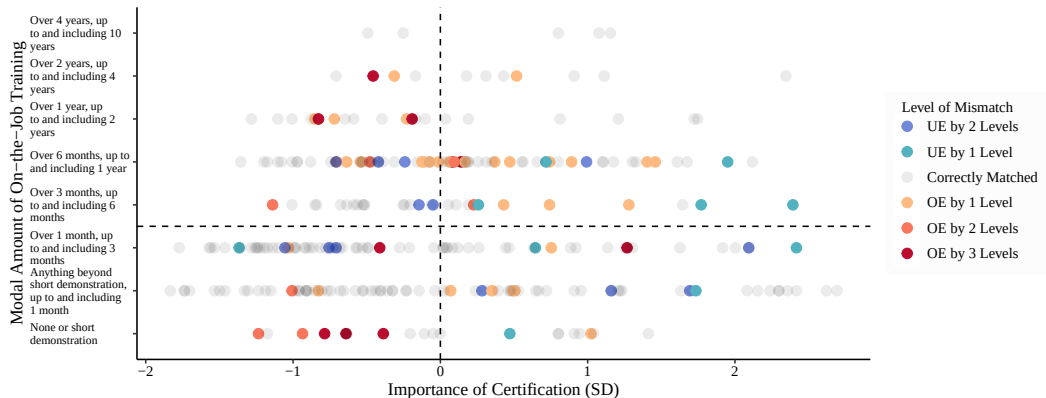
Notes: Level of mismatch is the difference between the modal degree of a non-citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

Figure A4: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



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Figure A5: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US

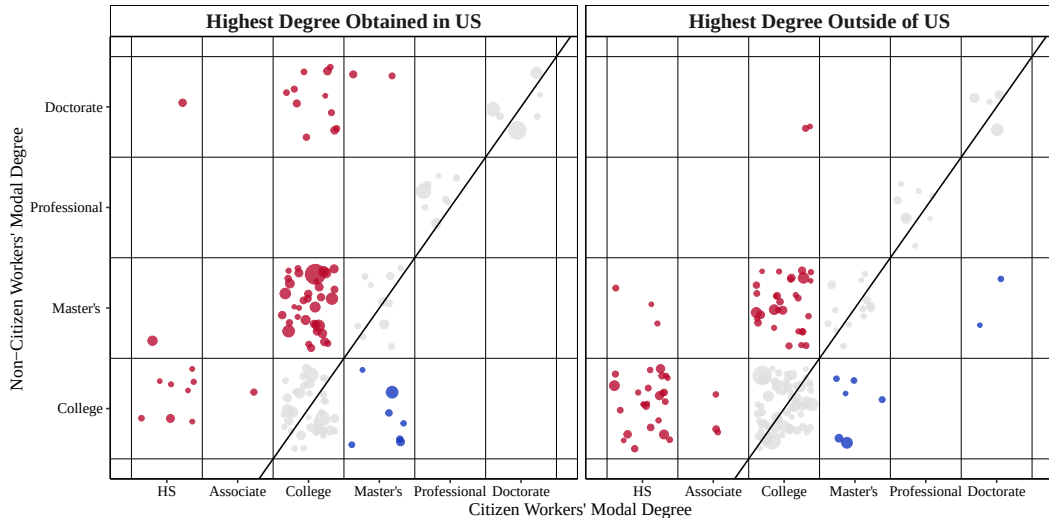


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Location of Highest Degree

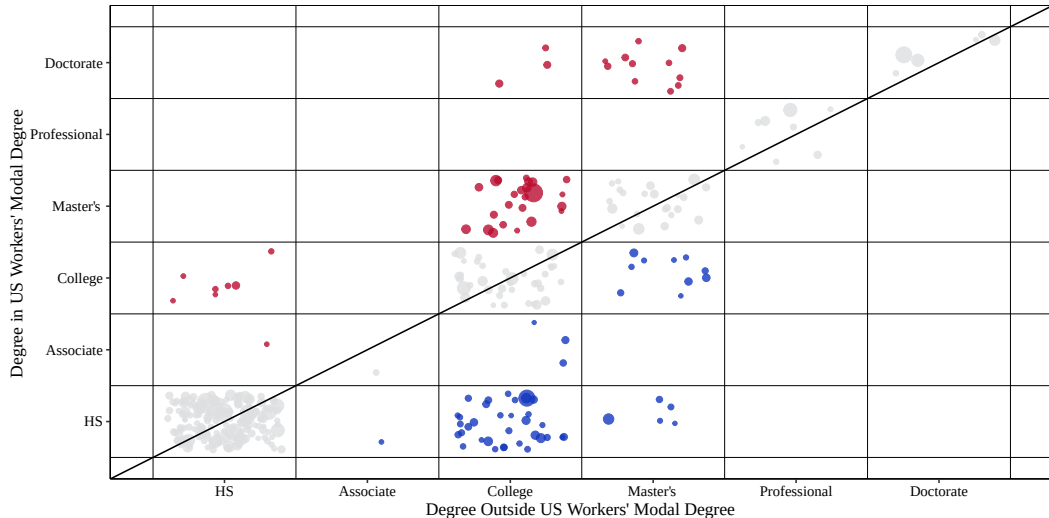
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Figure A6: Educational Mismatch of Non-Citizens, 2010-2019



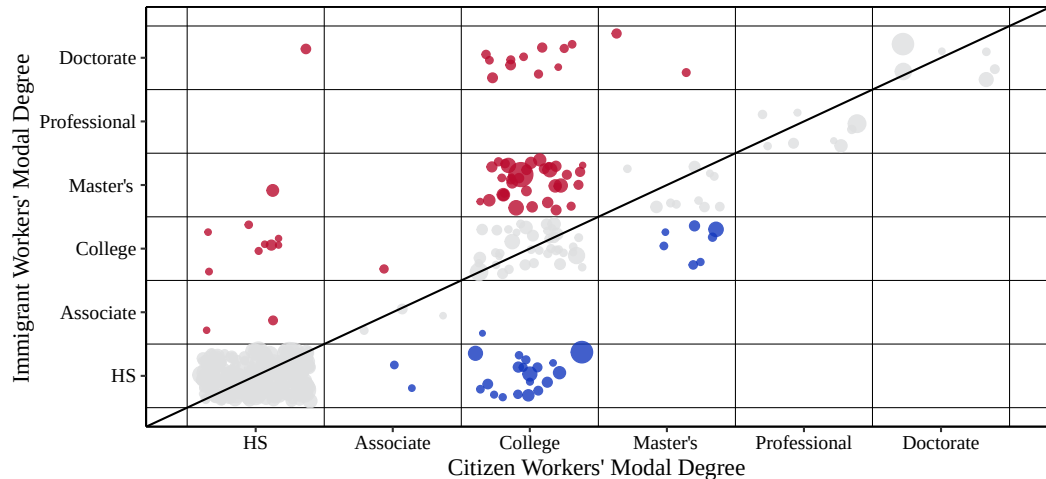
Location of Highest Degree [Back](#)

Figure A7: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree [Back](#)

Figure A8: Educational Mismatch of Non-Citizens, Highest Degree in U.S., 2010-2019



Location of Highest Degree [Back](#)

Figure A9: Educational Mismatch of Non-Citizens, Highest Degree Outside of U.S., 2010-2019

